Case 10: WUI Structural Resolution Sensitivity

1 Verification purpose

This case verifies sensitivity of firebrand-driven WUI spread to structural spatial representation. Five simulation conditions cover the baseline grid, native fine and coarse grids, and aligned and misaligned integrated-structure representations under one physical geometry and source definition.

The test quantifies native cell-scale distortion and determines whether integrated representation restores structure-level accumulation, ignition chronology, and spread-rate consistency. Capability-gated simulation conditions remain explicit and are not replaced by a physically different native-cell calculation.

2 Mathematical formulation and numerical implementation

2.1 Heat release and generation

For a structure ignited at t_i, the design heat-release-rate (HRR) per unit area (HRRPUA) is

$$q''(\tau) = \begin{cases} q''_{\max}\tau/300, & 0 < \tau < 300, \\ q''_{\max}, & 300 \le \tau \le 3900, \\ q''_{\max}(4200-\tau)/300, & 3900 < \tau < 4200, \\ 0, & \text{otherwise}, \end{cases} \qquad \tau = t - t_i, \qquad q''_{\max} = 400\,\mathrm{kW\,m^{-2}}.$$

For footprint A_s, $\mathrm{HRR}_i = q''A_s$, and the emission rate is

$$\dot{N}_i(t) = G'_R \mathrm{HRR}_i(t), \qquad G'_R = 10\,\mathrm{pcs\,MW^{-1}\,s^{-1}}.$$

The configuration therefore uses per-MW generation; per-area generation would not preserve heat-release-scaled production for this design fire.

2.2 Himoto transport and analytical load

The downwind distance is lognormal, with physical-space moments

$$\mu_x/D = 0.47(B^*)^{2/3}, \qquad \sigma_x/D = 0.88(B^*)^{1/3},$$

$$B^* = \frac{U}{\sqrt{gD}} \left(\frac{\rho_p}{\rho_\infty}\right)^{-3/4} \left(\frac{d_p}{D}\right)^{-3/4} \left(\frac{\mathrm{HRR}}{\rho_\infty c_p T_\infty g^{1/2} D^{5/2}}\right)^{1/2}.$$

Here $D = 10\,\mathrm{m}$, $\rho_p = 100\,\mathrm{kg\,m^{-3}}$, and $d_p = 5\,\mathrm{mm}$, matching the reference formulation and current source. The postprocessor independently evaluates the analytical deposited-load convolution

$$N_k = \sum_{i<k} \int_0^T \dot{N}_i(t) \left[\int_{x_k}^{x_{k+1}} f(x - x_i; \mu_i(t), \sigma_i(t))\, dx \right] dt$$

with a 5 s quadrature interval. Deposited counts are divided by physical footprint area, not a resolution-dependent raster-cell area.

2.3 Physical ignition and expected chronology

The configured ignition model uses $N_{\text{crit}} = 2928.9\,\text{pcs}\,\text{m}^{-2}$ [1]. At the structure spanning $20 \leq x \leq 30\,\text{m}$, the historical analytical load reaches this threshold near 2496.9 s. The case-specific postprocessor preserves the published composite full-ignition reference of 2858.8 s; its difference from direct addition of the currently configured delays is quantified explicitly below. Targets beyond the roughly 80 m peak-heat-release rate (HRR) spotting range require successive structure ignition and consequently have nonlinear arrival times.

2.4 Resolution signatures

On the 2.5 m native mesh (native fine-grid simulation condition), one physical structure occupies multiple independent cells. Uneven deposition and cell-local ignition should produce a nonzero within-structure load range and mean rate of spread (ROS) near $0.001\,\text{m}\,\text{s}^{-1}$, compared with about $0.006\,\text{m}\,\text{s}^{-1}$ for the exact-grid baseline. On the 30 m native mesh (native coarse-grid simulation condition), mode resampling inflates footprint and HRR, causing artificially early spread; the documented response gives a baseline-to-coarse load ratio near 1.86 [1].

The integrated-structure treatment groups all cells belonging to one structure,

$$\bar{N} = \frac{\sum_{j \in s} N_j}{A_s}, \qquad \text{HRR}_s = q'' A_s,$$

then redistributes load uniformly and emits from the structure centroid. The integrated-aligned simulation condition expects an aligned 2.5 m mesh to reproduce the baseline. The integrated-misaligned simulation condition permits up to 33% load deviation and 10% ignition-time deviation on a misaligned 3.24 m mesh.

3 Derivation of expected behavior

The exact grid should reproduce the analytical structure-load and ignition chronology. Native fine cells should expose within-structure imbalance, native coarse cells should expose footprint inflation, and integrated simulation conditions should recover the physical-structure response independently of alignment.

At the $400\,\text{kW}\,\text{m}^{-2}$ plateau, one physical 10×10 m structure has

$$HRR_{\text{max}} = 400(100) = 40000\ \text{kW} = 40\ \text{MW}, \qquad \dot{N}_{\text{max}} = 10(40) = 400\ \text{pcs}\,\text{s}^{-1},$$

and its critical total count is $N_{\text{crit}} = 2928.9(100) = 292890$ particles. Using the postprocessor's adjusted wind $U = 0.447(40)/0.87 = 20.5517\,\text{m}\,\text{s}^{-1}$ at peak HRR gives

$$B^* = 7.37372, \qquad \bar{x} = 0.47(7.37372)^{2/3}(10) = 17.8056\ \text{m},$$

$$s_x = 0.88(7.37372)^{1/3}(10) = 17.1282\ \text{m}, \qquad (\mu, \sigma) = (2.55195, 0.80939).$$

These values are inserted in the cell-cumulative distribution function (CDF) convolution at each 5-s midpoint; the HRR dependence is recomputed rather than freezing the peak values.

There is an explicit provenance discrepancy in the historical chronology. The case-specific constants give

$$2496.9 + 42.1 + 300 = 2839.0\ \text{s},$$

whereas the preserved historical composite reference used by postprocessing is 2858.8 s, 19.8 s later. The latter is therefore treated as a published numerical estimate, not as the arithmetic result of the three

displayed constants. The 19.8-s difference is only $19.8/2858.8 = 0.00693$, well inside the predeclared 10-percent baseline ignition tolerance, so this provenance mismatch cannot change that component's decision.

The exact-grid ideal errors are zero for analytical load and the chosen historical ignition reference. Native-grid expected signatures are a fine/baseline ROS ratio near $0.001/0.006 = 0.1667$ and a coarse load ratio near 1.86. Integrated aligned references are the exact-grid structure loads and times; for the misaligned grid, one unresolved boundary strip is $3.24/10 = 0.324$, motivating the explicit 0.33 load allowance.

4 Verification metrics and acceptance criteria

The postprocessor calculates every metric explicitly. The exact-grid baseline compares structure-integrated load with the analytical convolution defined above and second-structure ignition with 2858.8 s. The native fine and coarse simulation conditions verify the documented cell-local supporting results rather than claiming convergence. The integrated aligned and misaligned simulation conditions compare physical-structure profiles with the exact-grid baseline. Missing output or capability is reported as *NOT EVALUABLE*, never as an implicit pass.

4.1 Justification of metric quantification

The criteria intentionally separate reproduction of native-grid defects from accuracy requirements for the baseline and integrated treatments. The 10% baseline load and ignition bands accommodate rasterized deposition and threshold-crossing time while remaining small relative to the resolution effects documented for this configuration [1]. A native fine-grid load range of at least 5% is set above round-off and output quantization, and a ROS ratio no greater than 0.50 requires the material factor-of-two slowdown that identifies independent-cell behavior rather than a visually detectable but negligible difference. The coarse-grid load ratio is centered on the reported 1.86 response; its 25% band permits the severe footprint snapping intrinsic to a 30 m cell without accepting a reversal of the expected inflation.

For the aligned integrated grid, exact structure boundaries and conservation of integrated load justify the stringent 1% load limit. Its 5% ignition limit is wider because ignition is a nonlinear threshold event recorded at a discrete time. For the 3.24 m misaligned grid, one boundary-cell width relative to a 10 m structure is $3.24/10 = 0.324$; the 0.33 maximum load-error limit therefore admits approximately one unresolved boundary strip but not a larger structure-scale bias. The corresponding 10% ignition bound allows threshold amplification of that overlap error. Capability and completeness are conjunctive because the integrated claims cannot be inferred from native-grid substitutes.

4.2 Predeclared acceptance criteria

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Baseline load error	Mean structure-load relative error against the analytical deposition reference, excluding the source structure.	≤ 0.10
Baseline ignition error	Relative error of second-structure ignition time against 2858.8 s.	≤ 0.10
Native fine-grid load distortion	Median within-structure relative load range.	≥ 0.05
Native fine-grid ROS ratio	Fine-grid fitted mean ROS divided by baseline ROS.	≤ 0.50
Native coarse-grid load ratio	Baseline structure-4 load divided by coarse structure-2 load.	$1.86 \pm 25\%$
Integrated aligned load error	Mean matched structure-load relative error against baseline.	≤ 0.01
Integrated aligned ignition error	Mean matched ignition-time relative error against baseline.	≤ 0.05
Integrated misaligned load error	Maximum matched structure-load relative error against baseline.	≤ 0.33
Integrated misaligned ignition error	Maximum matched ignition-time relative error against baseline.	≤ 0.10
Overall decision	All configured simulation conditions and metrics must be evaluable; unavailable integrated capability prevents a PASS.	PASS only if all rows pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for the 10 m baseline grid. These panels show the prepared spatial fields, remove the two-cell numerical buffer, and retain map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

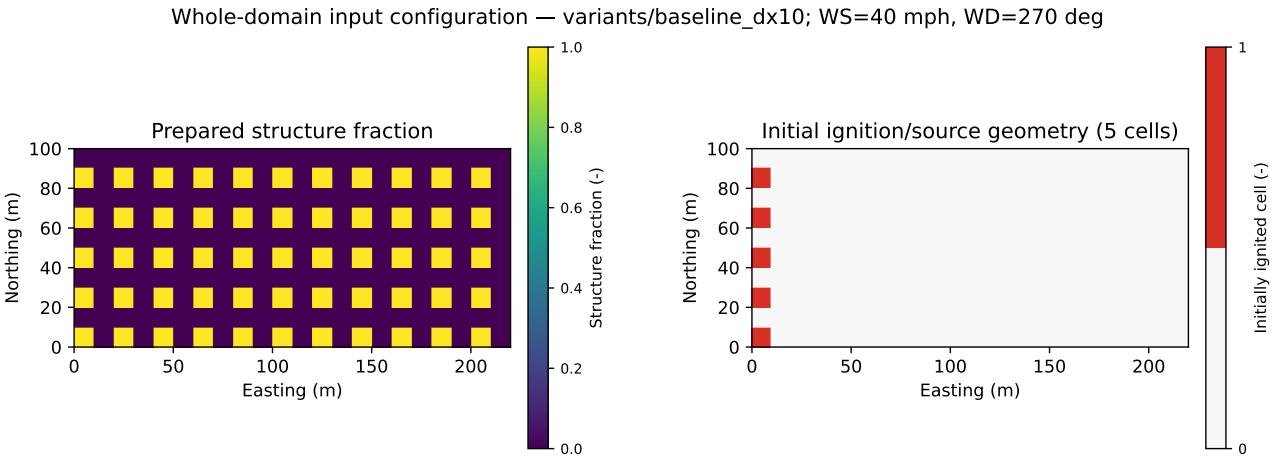


Figure 1: Whole-domain prepared input configuration for the representative simulation condition. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

Table 2: Five simulation conditions generated from the common physical problem.

Simulation condition	Δx (m)	$t_{\max}$ (s)	Treatment	Expected result
Baseline	10.00	22000	native	analytical deposition and ignition
Native fine	2.50	50000	native fine	imbalance and slower ROS
Native coarse	30.00	22000	native coarse	inflated HRR and earlier spread
Integrated aligned	2.50	22000	integrated aligned	baseline-equivalent profile
Integrated misaligned	3.24	22000	integrated misaligned	bounded profile error

Each nominal timestep is $\Delta t_{\text{nominal}} = 0.5\Delta x/U$; the generated step is no larger than this bound. All structure rasters reference a case-specific model 14 containing the design fire and physical delay. Structures form a periodic 220×100-m two-dimensional lattice. Each represented cell uses its true square area, so heat-release rate per unit area (HRRPUA) remains 400 kW m^{-2} and the dimensionless ignition-hardening factor remains one at every resolution. Every structure in the first upwind column is initialized at $t = 0$. The global vegetative surface-spread setting switch remains true to suppress vegetative surface-fire propagation. In the current source, initialized model-14 buildings independently refresh their transient HRR and therefore remain active under this setting. The generated non-structure fuel model 93 is also nonburnable, providing a second explicit isolation of the building-only experiment. Because the wind Courant–Friedrichs–Lewy (CFL) bound requires a subsecond step, preprocessing computes the nominal fractional value. When that value does not divide the requested duration, it uses $N = \lceil t_{\text{requested}}/\Delta t_{\text{nominal}} \rceil$ and reduces the generated step to $\Delta t = t_{\text{requested}}/N$. The stop therefore remains the exact requested duration and an exact timestep multiple; the simulation specification records nominal and actual steps, CFL values, and the integer step count.

5.1 Preprocessing, execution, and postprocessing

Exact cell–footprint overlap defines the structure fractions and identities; mode classification defines the native fine- and coarse-grid experiments. Only model-supported structure representations are simulated. Final firebrand counts are normalized by physical structure area and aggregated by structure identity, together with the fire arrival times. Evaluation requires agreement between the prescribed and simulated inputs, successful termination, unique stop-time, timestep, and meteorological-interval values, and exact divisibility of duration by timestep. One final output-time record and one corresponding deposition/arrival-time pair are required. An output beyond the nominal stop is accepted only if subtracting the meteorological interval gives a time on the timestep grid within one timestep of the stop. Ambiguous, early, off-grid, or duplicate evidence remains NOT EVALUABLE.

6 Actual simulation results

Figure 2 shows the representative time-of-arrival field from the 10 m baseline grid. The displayed field is taken from the simulated results, masks missing values, and excludes the two-cell numerical buffer. This completed baseline provides whole-domain evidence even though simulation conditions requiring the unavailable integrated-structure capability remain not evaluable.

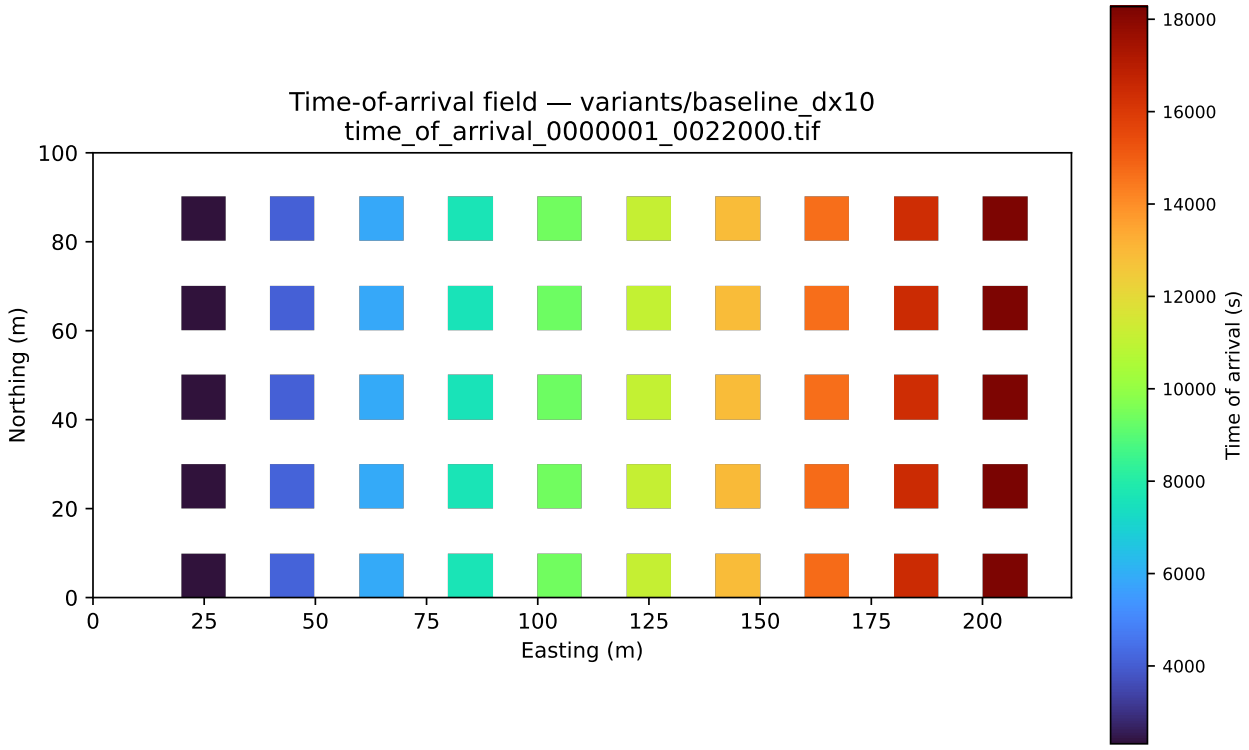


Figure 2: Whole-domain time-of-arrival field from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

2 of 5 simulation conditions currently have complete, configuration-matched ELMFIRE output. Source-aligned physical-state figures are provided for all three native representations; a figure explicitly states when its current matching output is unavailable. The two integrated-structure simulation conditions remain capability gated.

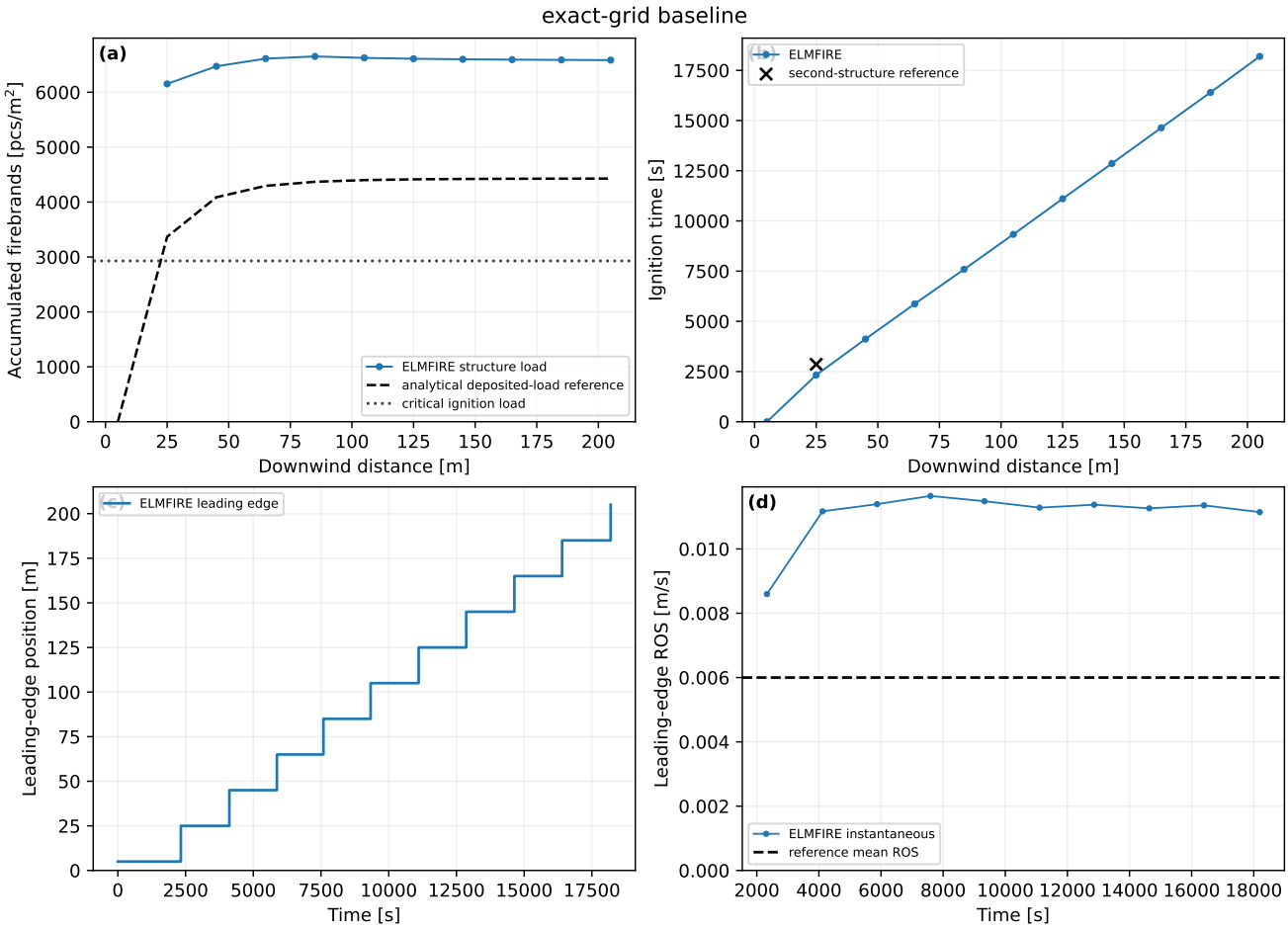


Figure 3: Exact-grid baseline observables: structure load, ignition chronology, leading-edge trajectory, and instantaneous ROS. Analytical load and second-structure timing references are overlaid where available.

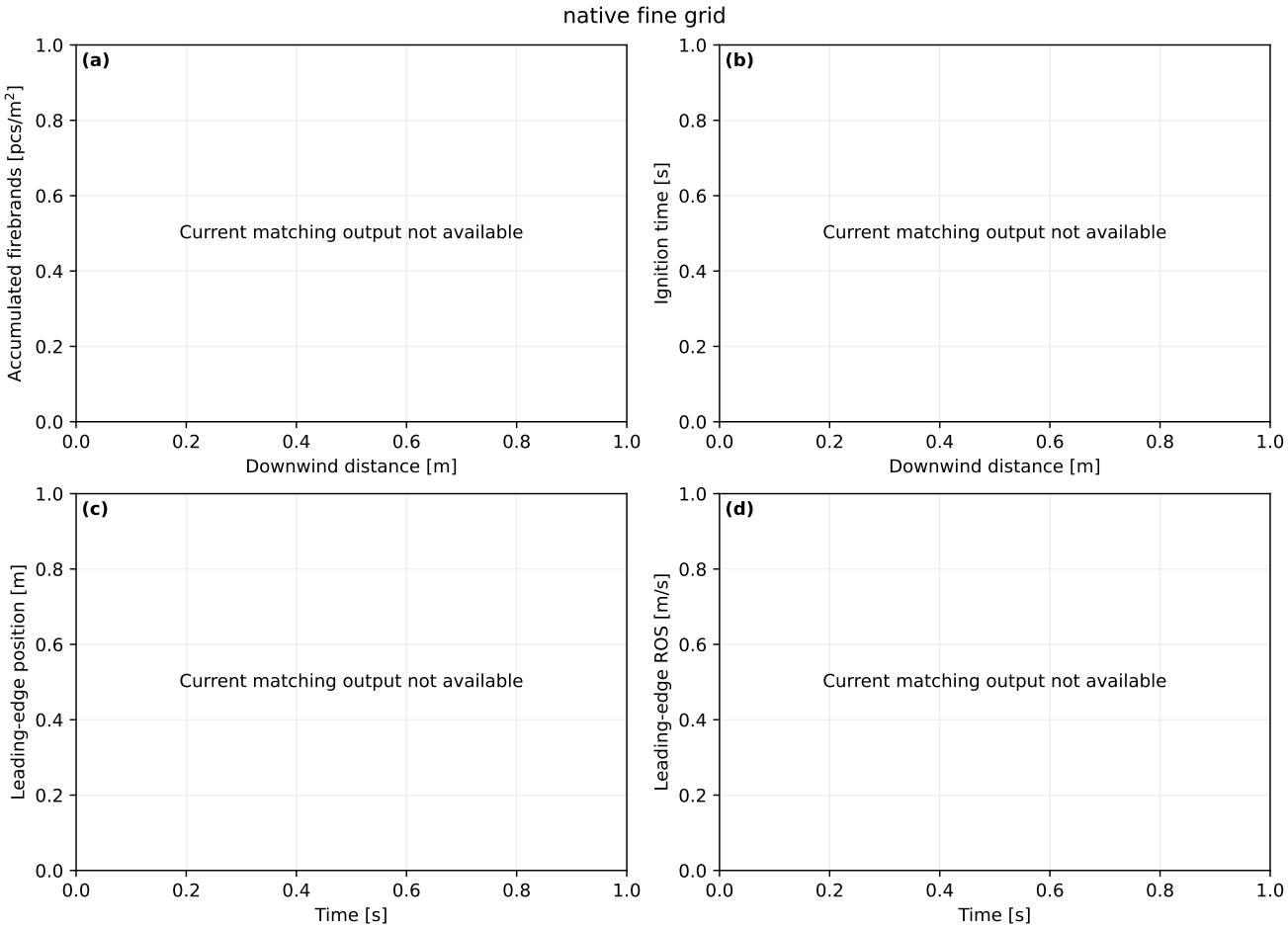


Figure 4: Native fine-grid observables. Cell-scale load variation within each physical structure is retained so discretization distortion is visible rather than hidden by structure averaging.

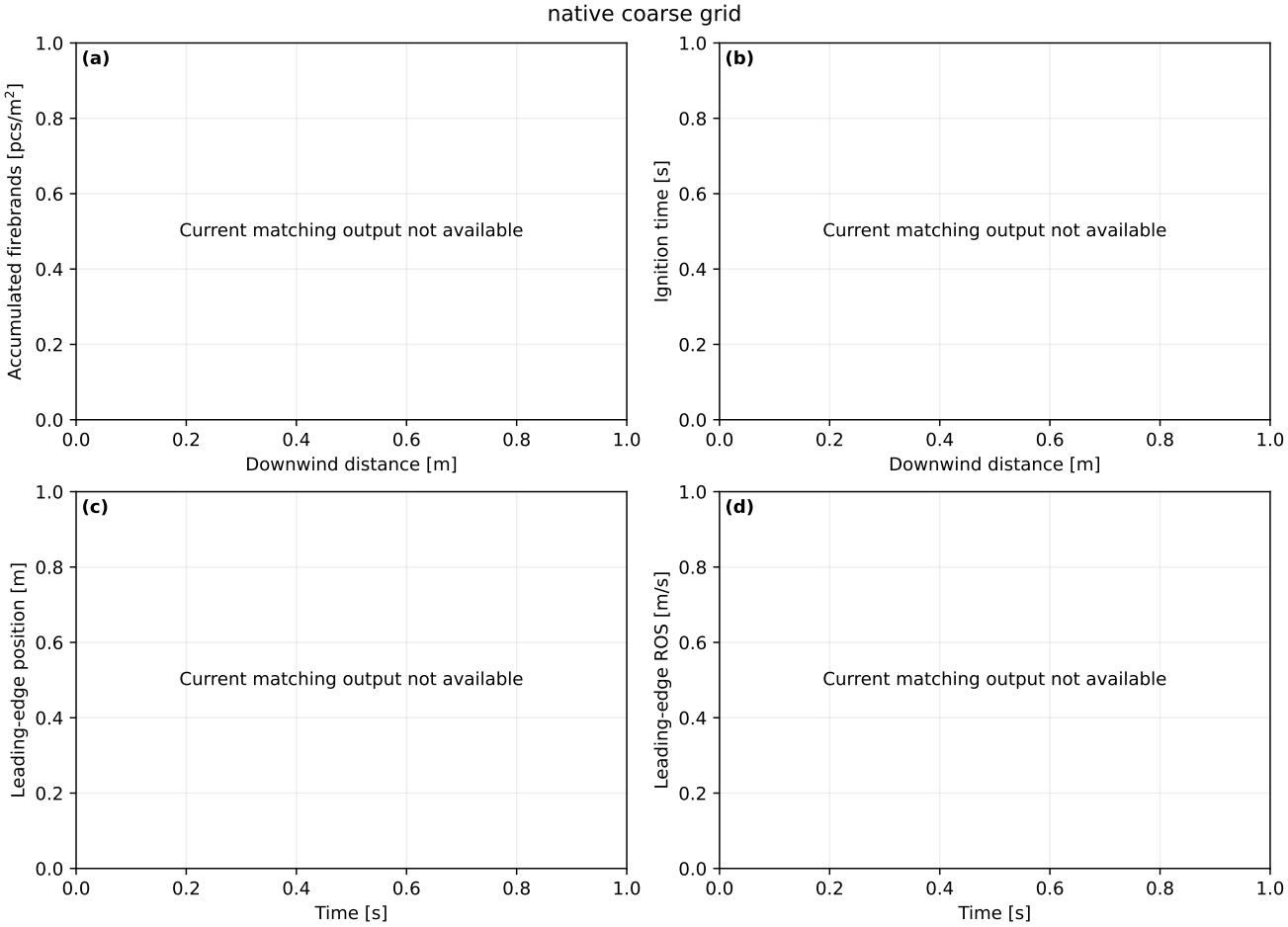


Figure 5: Native coarse-grid observables. Mode resampling changes the represented footprint and separation, and the resulting load and ignition histories are shown on their actual physical coordinates.

Figure 6 is the supplemental cross-simulation condition diagnostic. The native-test decision is **FAIL**; the complete-suite decision is **NOT EVALUABLE (capability missing)**.

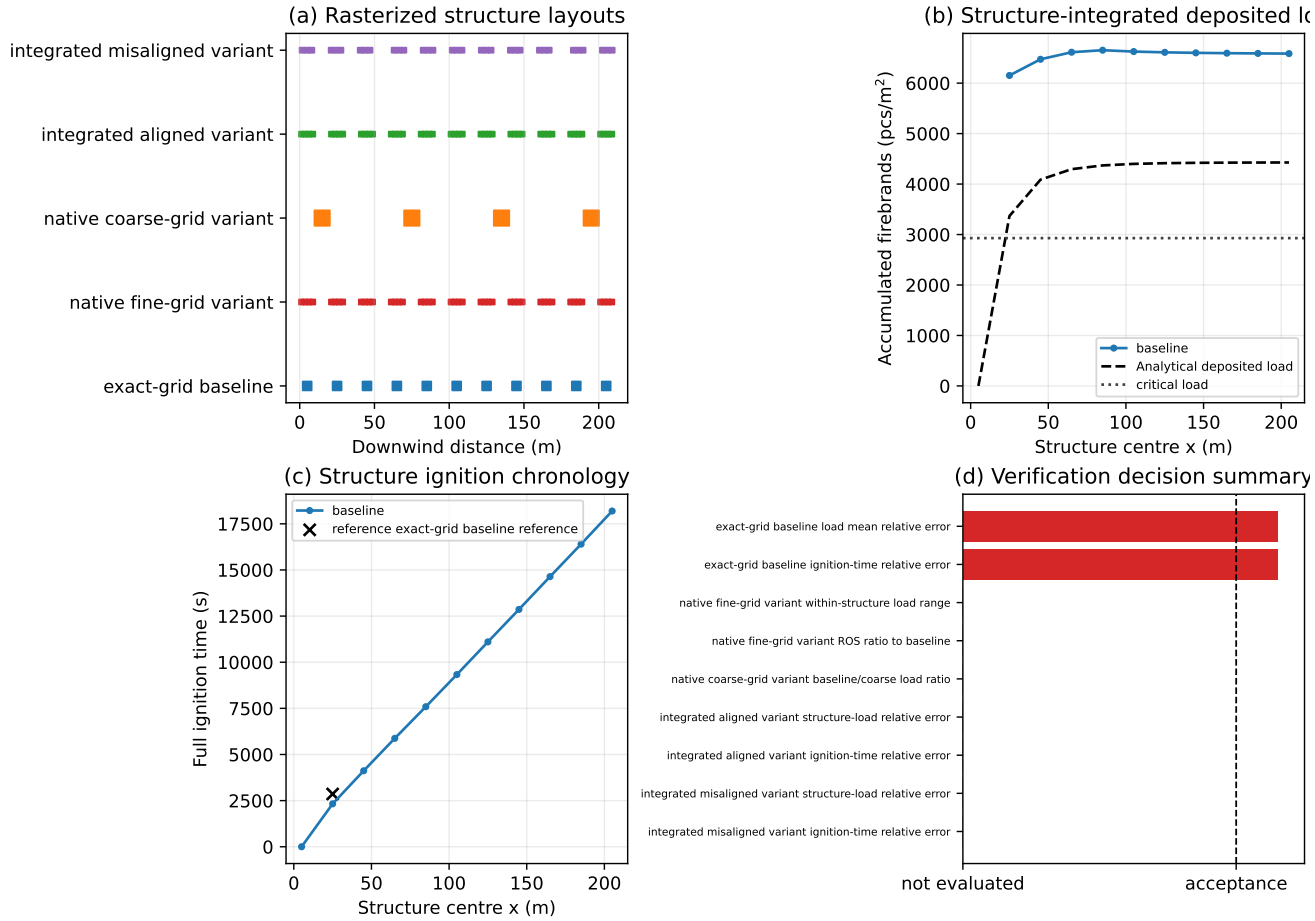


Figure 6: Structural-resolution summary. (a) Rasterized layouts; (b) deposited load by physical structure; (c) ignition chronology; and (d) component decision status, with unsupported components shown as not evaluated.

7 Calculated metrics and verification decision

Table 3: Calculated verification metrics and component decisions.

Metric	Acceptance limit	Calculated	Status
exact-grid baseline load mean relative error	≤ 0.1	0.5435	FAIL
exact-grid baseline ignition-time relative error	≤ 0.1	0.1861	FAIL
native fine-grid simulation condition within-structure load range	≥ 0.05	N/A	NOT EVALUABLE
native fine-grid simulation condition ROS ratio to baseline	≤ 0.5	N/A	NOT EVALUABLE
native coarse-grid simulation condition baseline/coarse load ratio	1.86 +/- 25%	1.622	PASS
integrated aligned simulation condition structure-load relative error	≤ 0.01	N/A	NOT EVALUABLE
integrated aligned simulation condition ignition-time relative error	≤ 0.05	N/A	NOT EVALUABLE
integrated misaligned simulation condition structure-load relative error	≤ 0.33	N/A	NOT EVALUABLE
integrated misaligned simulation condition ignition-time relative error	≤ 0.1	N/A	NOT EVALUABLE

The cited model description [1] states that the integrated-structure method had not yet been implemented in ELMFIRE. The preprocessor confirms this against the current source. The integrated aligned and misaligned cases are therefore generated as prospective simulation conditions but are not run while that feature is absent. Substituting the native cell-local formulation would change the prescribed experiment and could produce misleading conclusions about resolution independence. Earlier input configurations are excluded from the present comparison.

8 Technical interpretation

The corrected exact-grid baseline and native coarse simulation condition both terminate at an exact 22 000 s final dump with matching input identity records. The baseline load error is 0.5435 and its ignition-time error is 0.1861, both above the 0.10 limits; these are genuine numerical failures. The coarse baseline-to-coarse load ratio is 1.622 and passes the prescribed $1.86 \pm 25\%$ band.

The native fine-grid member was executed toward its prescribed 50 000 s endpoint but could not complete safely. It reached 36 000.065 s; resident memory grew from approximately 204 MB near 15 000 s to 960 MB, while successive 500 s dump intervals slowed from roughly 198 wall seconds to about 11 minutes. The run was stopped before exhausting the shared environment, and no completion input identity record was written, so its partial rasters are correctly excluded. This reproduces an implementation resource-scaling limitation in the active ember-tracking path, not a stop-time or raster-configuration defect.

The integrated simulation conditions test whether one physical structure remains the source and collector; their missing capability is a source-interface limitation and is likewise reported as not evaluable rather than as a numerical pass or failure. Consequently the native-test status is **FAIL**, while the full suite remains **NOT EVALUABLE (capability missing)**.

9 Scope and limitations

The case is an idealized two-dimensional WUI lattice with fixed structure geometry and wind. A representative cross-stream lattice row supplies the one-dimensional structure profiles, while whole-domain maps expose unintended lateral behavior. The case verifies resolution behavior and capability coverage, not the physical accuracy of the structural transport or ignition correlations.

References

[1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.